

DATA ANALYTICS PROJECT REPORT

HR Workforce & Attrition Analytics

End-to-End Employee Attrition Analysis

Python · PostgreSQL · SQL · Power BI

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1,470 Employees Analyzed • 237 Departures • 16.12% Attrition Rate

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1. Project Overview

This project is an end-to-end data analytics case study built on an internal HR workforce dataset, covering the full pipeline from raw data to business-ready insight. The workflow spans data loading and understanding, data cleaning, exploratory data analysis (EDA), feature engineering, relational database integration, SQL-based business analysis, and an interactive Power BI dashboard.

The objective was to understand where and why employee attrition is concentrated across the organization — by department, role, compensation, tenure, workload, and satisfaction — and to package those observations into a dashboard and set of recommendations that a People/HR team could act on.

2. Business Problem

Employee attrition is costly: it disrupts teams, increases hiring and onboarding overhead, and can signal deeper issues in compensation, workload, or employee experience. Without structured analysis, HR teams often rely on anecdotal impressions rather than data when deciding where to focus retention efforts.

This project addresses that gap by quantifying attrition across multiple workforce dimensions — department, job role, salary level, overtime status, tenure, job satisfaction, environment satisfaction, and commute distance — so that retention efforts can be directed toward the segments where departures are most concentrated.

3. Objectives

- Clean and standardize a raw HR dataset, resolving duplicate and conflicting records.
- Perform exploratory data analysis to quantify attrition across key workforce dimensions.
- Engineer segment-level features such as tenure, distance, and promotion groups to support grouped analysis.
- Load the cleaned dataset into PostgreSQL and answer targeted business questions using SQL.
- Design an interactive Power BI dashboard for ongoing, self-serve workforce monitoring.
- Translate observed patterns into non-causal, actionable business recommendations.

4. Dataset Overview

The raw dataset contained 1,480 employee records across 38 columns. After deduplication and cleanup, the final analytical dataset consists of 1,470 unique employees across 37 columns.

Metric	Value
Original records	1,480
Original columns	38
Final unique employees	1,470
Final columns	37
Exact duplicate rows removed	7
Duplicate employee IDs identified	10
Remaining missing values	60 (years_with_curr_manager)

5. Tools & Technologies

Tool	Role in Project
Python (Pandas)	Data loading, cleaning, EDA, and feature engineering
Jupyter Notebook	Interactive, documented analysis workflow
PostgreSQL	Relational storage of the cleaned dataset
SQL	Structured business queries against the workforce data
Power BI	Interactive dashboard for stakeholder-facing reporting

6. Data Cleaning & Preparation

The raw dataset required deduplication and light standardization before analysis:

- Removed 7 exact duplicate rows.
- Identified 10 duplicate employee IDs; consolidated the duplicate records.
- 3 of those employee IDs had conflicting years_with_curr_manager values — rather than guessing, these were treated as missing and the records were merged.
- Removed columns with no analytical value: EmployeeCount, Over18, StandardHours, EmployeeNumber.
- Standardized all column names to snake_case for consistency across Python, SQL, and Power BI.
- The only missing values remaining after cleaning were 60 values in years_with_curr_manager, which were kept as true nulls rather than imputed or replaced with placeholder text.

Result: a clean, analysis-ready dataset of 1,470 rows × 37 columns.

7. Exploratory Data Analysis

Headline workforce metrics from the cleaned dataset:

Metric	Value
Total employees	1,470
Employees who left	237
Employees who stayed	1,233
Overall attrition rate	16.12%
Average age	36.92 years
Average monthly income	6,502.93
Average years at company	7.01

Attrition by Department

Department	Attrition Rate
Sales	20.63%
Human Resources	19.05%
Research & Development	13.84%

Attrition by Salary Level

Salary Slab	Attrition Rate
Up to 5k	~21.8%
5k–10k	~11.2%
10k–15k	~13.5%
15k+	~3.8%

A clear inverse relationship is observed between salary level and attrition — the lowest salary slab shows roughly 5× the attrition rate of the highest slab.

Attrition by Overtime Status

Overtime	Attrition Rate
Yes	~30.53%
No	~10.44%

Attrition by Job Satisfaction

Satisfaction Level	Attrition Rate
1 (Low)	22.9%
2	16.4%
3	16.5%
4 (High)	11.3%

Attrition Across Tenure Stages

Tenure	Attrition Rate
0–2 years	29.82%
3–5 years	13.82%
6–10 years	12.28%
11–15 years	6.48%
16+ years	9.42%

Attrition is heavily concentrated in the earliest tenure band — employees in their first two years leave at more than 3× the rate of the 11–15 year band.

Attrition by Distance from Home

Distance from Home	Attrition Rate
0–5	~13.8%
6–10	~14.5%
11–20	~20.0%
21+	~22.1%

Attrition by Environment Satisfaction

Satisfaction Level	Attrition Rate
1 (Low)	~25.4%
2	~15.0%
3	~13.7%
4 (High)	~13.4%

8. Feature Engineering

To support grouped and dashboard-level analysis, several categorical features were derived from continuous fields:

- tenure_group — years_at_company bucketed into 0–2, 3–5, 6–10, 11–15, and 16+ year bands.
- salary_slab — monthly_income bucketed into Up to 5k, 5k–10k, 10k–15k, and 15k+ bands.
- age_group — age bucketed into standard bands (18–25, 26–35, 36–45, 46–55, 55+) used in the dashboard's departure breakdown.

These engineered features made it possible to analyze attrition by segment rather than by raw continuous values, which is what both the SQL queries and the Power BI visuals rely on.

9. PostgreSQL & SQL Analysis

The cleaned dataset was loaded into a PostgreSQL database. Nine SQL queries were then written to answer specific, stakeholder-relevant questions about where attrition is concentrated.

Query 1 — Overtime-related attrition exposure by department

Department	Overtime Employees	Overtime Departures	Overtime Attrition Rate
Sales	128	48	37.50%
Human Resources	17	5	29.41%
Research & Development	271	74	27.31%

Query 2 — Attrition concentration by department and job role

Department	Job Role	Employees	Left	Rate
Sales	Sales Representative	83	33	39.76%
R&D	Laboratory Technician	259	62	23.94%
Human Resources	Human Resources	52	12	23.08%
Sales	Sales Executive	326	57	17.48%
R&D	Research Scientist	292	47	16.10%
R&D	Manufacturing Director	145	10	6.90%

Full results cover all 11 department × role combinations in the dataset; the table above highlights the highest and lowest ends of the range.

Query 3 — Attrition across salary levels

Salary Slab	Employees	Left	Attrition Rate
Upto 5k	749	163	21.76%
5k–10k	440	49	11.14%
10k–15k	148	20	13.51%
15k+	133	5	3.76%

Query 4 — Job-role share of total departures

Job Role	Employees Left	Share of Departures	Rank
Laboratory Technician	62	26.16%	1
Sales Executive	57	24.05%	2
Research Scientist	47	19.83%	3
Sales Representative	33	13.92%	4
Human Resources	12	5.06%	5
Manufacturing Director	10	4.22%	6
Healthcare Representative	9	3.80%	7
Manager	5	2.11%	8
Research Director	2	0.84%	9

Query 5 — Employee profile: left vs. stayed

Group	Employees	Avg. Age	Avg. Income	Avg. Tenure	Avg. Distance	Avg. Job Satisfaction
Stayed	1,233	37.6	6,832.74	7.37	8.92	2.78
Left	237	33.6	4,787.09	5.13	10.63	2.47

On average, employees who left were younger, earned less, had shorter tenure, commuted farther, and reported lower job satisfaction than those who stayed.

Query 6 — Attrition by job satisfaction

Job Satisfaction	Employees	Left	Attrition Rate
1	289	66	22.84%
2	280	46	16.43%
3	442	73	16.52%
4	459	52	11.33%

Query 7 — Employees Matching Key Attrition Indicators

This query identifies employees who had already left and matched three observed attrition indicators: early tenure, overtime, and low-to-moderate job satisfaction. A sample of the flagged records:

Emp ID	Department	Job Role	Yrs at Co.	Satisfaction	Overtime	Income
RM458	Sales	Sales Representative	0	2	Yes	1,878
RM586	R&D	Laboratory Technician	0	1	Yes	1,601
RM1061	R&D	Laboratory Technician	0	1	Yes	3,172
RM712	R&D	Research Scientist	0	1	Yes	2,404
RM1256	Sales	Sales Executive	0	1	Yes	8,564
RM1340	R&D	Research Scientist	1	2	Yes	2,472
RM1111	R&D	Laboratory Technician	1	1	Yes	2,074

Emp ID	Department	Job Role	Yrs at Co.	Satisfaction	Overtime	Income
RM205	R&D	Healthcare Rep.	1	1	Yes	6,673

Query 8 — Attrition across tenure stages

Tenure Group	Employees	Left	Attrition Rate
0–2 years	342	102	29.82%
3–5 years	434	60	13.82%
6–10 years	448	55	12.28%
16+ years	138	13	9.42%
11–15 years	108	7	6.48%

Query 9 — Attrition across overtime × job satisfaction

Overtime	Job Satisfaction	Employees	Left	Attrition Rate
Yes	2	69	26	37.68%
Yes	1	84	30	35.71%
Yes	3	121	41	33.88%
Yes	4	142	30	21.13%
No	1	205	36	17.56%
No	3	321	32	9.97%
No	2	211	20	9.48%
No	4	317	22	6.94%

The combination of overtime and low job satisfaction compounds risk: every overtime row shows a substantially higher attrition rate than its non-overtime counterpart at the same satisfaction level.

10. Power BI Dashboard

The analysis was consolidated into an interactive, two-page Power BI dashboard for ongoing workforce monitoring, with slicers for Department, Job Role, and Overtime status.

Page 1 — Workforce Overview

KPI cards for Total Employees, Employees Left, Attrition Rate, Average Tenure, and Average Monthly Income, alongside attrition breakdowns by department, salary level, overtime status, job role, and age group.

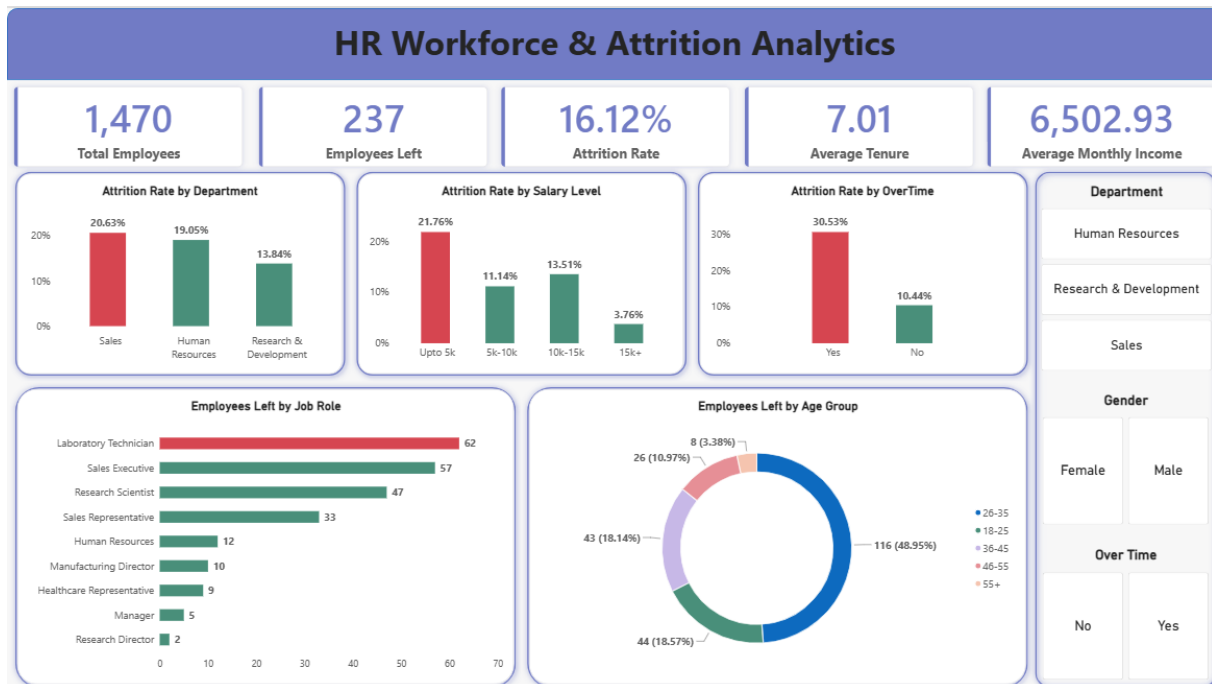


Figure 1 — Workforce overview: KPIs and departure breakdowns by department, salary, overtime, role, and age.

Page 2 — Attrition Drivers & Workforce Insights

A deeper drill-down page covering attrition across tenure stages, the overtime × job satisfaction attrition matrix, attrition by distance from home and job satisfaction, and early-tenure/overtime/low-satisfaction departures by job role.

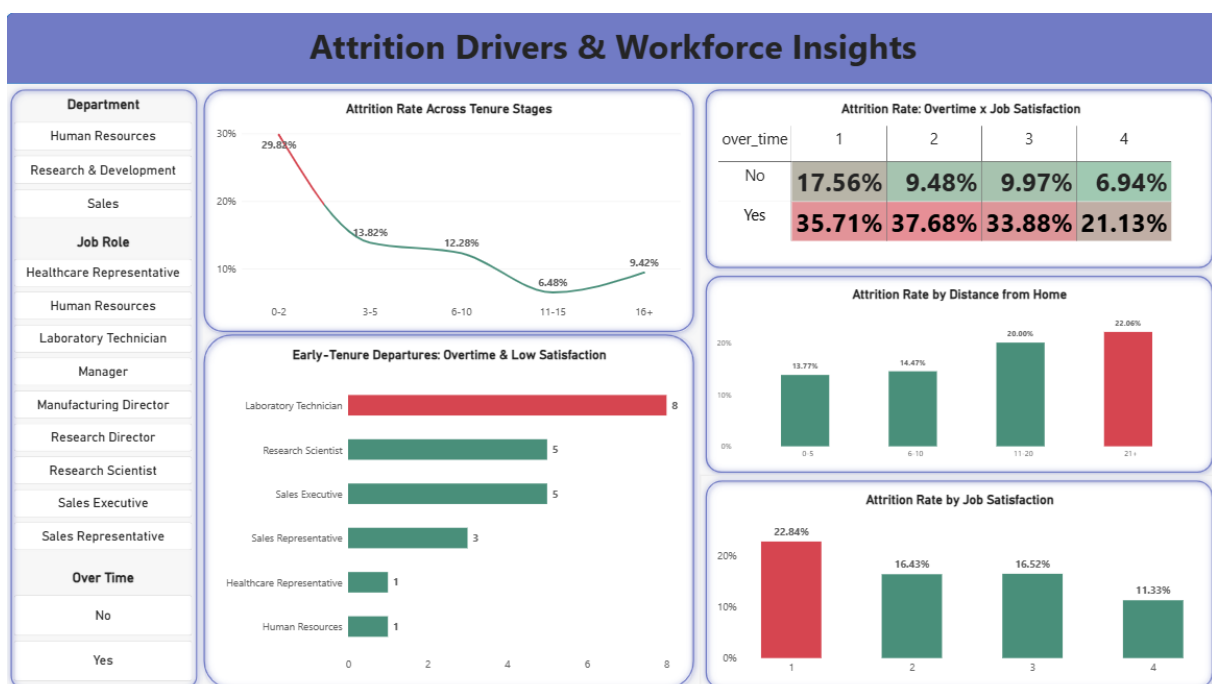


Figure 2 — Attrition drivers: tenure stages, overtime × satisfaction matrix, distance, and role-level risk.

11. Key Findings

- Overtime shows the strongest observed association with attrition: employees working overtime leave at roughly 3× the rate of those who do not (30.53% vs. 10.44%).
- Attrition is heavily front-loaded in tenure: the 0–2 year band shows a 29.82% attrition rate, well above every later tenure stage.
- Sales has the highest departmental attrition rate (20.63%), and the Sales Representative role has the highest role-level rate of any role (39.76%).
- Compensation shows a strong inverse association with attrition: the lowest salary slab (up to 5k) is associated with roughly 5× the attrition rate of the top slab (15k+).
- Job satisfaction and environment satisfaction both show an inverse pattern with attrition — lower reported satisfaction is associated with higher observed departure rates.
- Longer commute distance is associated with higher attrition, rising from ~13.8% (0–5 km) to ~22.1% (21+ km).
- Laboratory Technicians account for the single largest share of total departures (26.16%), despite the R&D department's overall attrition rate being the lowest of the three departments — an area for further investigation.
- Overtime and low job satisfaction compound: every overtime segment shows higher attrition than the equivalent non-overtime segment at the same satisfaction level.

These are observed statistical patterns in the dataset, not demonstrated causes of attrition; each is flagged here as an area for further investigation rather than a conclusion.

12. Business Recommendations

- Review the early-tenure employee experience (onboarding, early manager check-ins, ramp-up support), given the concentration of departures in the 0–2 year band.
- Investigate workload and overtime patterns, particularly in Sales and R&D, where overtime departures are highest in absolute terms.
- Review compensation patterns in the lowest salary slab and in roles with disproportionately high attrition, such as Sales Representative.
- Monitor low job-satisfaction and low environment-satisfaction segments as an early-warning signal rather than waiting for exit interviews.
- Examine the employee experience for staff with longer commuting distances (21+ km), where attrition is highest.
- Use the Power BI dashboard and its KPIs for ongoing, self-serve workforce monitoring rather than one-off analysis.

These are suggested areas for further HR investigation based on observed patterns, not prescriptions guaranteed to reduce attrition.

13. Limitations

- The analysis is observational and associative — it identifies patterns, not proven causes, of attrition.
- 60 missing values remain in years_with_curr_manager, which slightly limits manager-tenure-based analysis.

- The dataset is a single snapshot; it does not capture seasonal, year-over-year, or economic-cycle trends in attrition.
- Some role-level segments (e.g., Research Director, n = 80 employees, 2 departures) have small sample sizes, so their attrition rates should be read with caution.
- The dataset does not include qualitative context (e.g., exit interview reasons), which would help validate the drivers behind the observed patterns.

14. Conclusion

This project demonstrates a complete, practical data analytics workflow — from raw data cleaning through EDA, feature engineering, relational database integration, SQL-based business querying, and interactive dashboard design. The analysis identifies clear, data-backed patterns in workforce attrition, most notably around overtime, early tenure, compensation, and job satisfaction, and translates them into a dashboard and a set of investigation-focused recommendations that a People/HR team could use for ongoing monitoring.

15. Future Scope

- Extend the SQL layer with a scheduled ETL process so the PostgreSQL tables and dashboard refresh automatically as new HR data arrives.
- Incorporate additional data sources, such as exit-interview text or engagement-survey scores, to add qualitative context to the observed patterns.
- Explore predictive modeling as a separate follow-on project to estimate individual attrition risk, once a validated ground-truth labeling process is in place.
- Add row-level security and role-based views in Power BI so different stakeholders (HR, department heads) see the slice of data relevant to them.

16. Technical Skills Demonstrated

Area	Skills
Data Cleaning	Deduplication, conflict resolution, missing-value handling, schema standardization (Pandas)
Exploratory Analysis	Group-by aggregation, segment-level attrition analysis, descriptive statistics
Feature Engineering	Binning continuous variables into business-relevant categorical segments
Databases & SQL	PostgreSQL integration, business queries, aggregation, ranking
Data Visualization	Power BI dashboard design, KPI cards, interactive slicers, matrix visuals
Data Storytelling	Translating statistical patterns into clear, non-causal business recommendations